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Semantic Representations from LLMs for News Recommendation: A Controlled Study of 2024–2026 Directions on MIND

HUST

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Hà Nội, 2026

ONE LOVE. ONE FUTURE.

Executive Summary

82–86%
of test clicks hit articles never seen in training (MIND-small / large)

0.4094 vs 0.4078
best zero-shot LLM config vs retrained NRMS (MIND-small nDCG@10; CI includes 0)

+0.0182
category text in Qwen3-Embedding: the largest robust LLM-side gain

0.038 → 0.007
stacker validation optimism: user folds → time-block folds

- **Frozen LLM embeddings match — but do not beat — properly retrained NRMS / NAML / Fastformer;** robust gains come from history aggregation (late interaction, +0.0128) and input fields (+0.0182), not from encoder size.
- **Most 2024–2026 LLM-rec directions we tested give ≤ 0.005 or negative effects** under strict evaluation: representation-level RAG / GraphRAG, generative semantic IDs, a 1.7B LLM judge, multi-interest models, LoRA, an LLM user tower.
- **A warm validation day misleads selection:** 18.1% cold clicks vs 82.2% at test; RAG / GraphRAG flip sign and learned readers are inflated.
- **MIND-large:** linear fusion 0.6863 AUC / 0.4373 nDCG@10; GBDT helps only with label-free exposure (0.7013 / 0.4516) and saturates; the ≈ 0.02 AUC gap to PLM-NR points to end-to-end fine-tuning → gated v17 run (pending).

1. Problem Formulation & Measured Challenges

Task: impression (slate) ranking [Wu et al., ACL'20]

- User u with click history H = (h1 … hL); an impression shows candidates C = (c1 … cJ) with click labels y ∈ {0, 1}.
- Learn a score s(u, c); report AUC, MRR, nDCG@5 and nDCG@10 per impression, averaged (official MIND protocol).

Challenge 1 — item cold start dominates (measured)

- 85.9% (MIND-small) and 82.2% (MIND-large) of test clicks are on articles absent from every training row → ID-only collaborative filtering has no signal for most clicks; content representations are mandatory.

Challenge 2 — chronological shift between validation and test (measured)

- Validation = last training day: only 18.1% of its clicks are cold (MIND-large) → components that excel on seen articles look better than they are.

Challenge 3 — short text, no article body

- Titles 10.7 words, abstracts 34.9 words (5.2% empty); MIND ships URLs, not bodies → which fields reach the encoder matters (Section 4.3).

** MINDsmall_dev: 73,152 and MINDlarge_dev: 376,471 test impressions; every number in this deck is measured on these sets.*

2. Research Questions & Contributions

Research questions

- **RQ1** Where do zero-shot embeddings from modern encoders / LLMs stand against properly trained NRMS, NAML, Fastformer under one chronological protocol?
- **RQ2** Is the bottleneck the encoder (size, family), the input fields, or history aggregation?
- **RQ3** Do 2024–2026 LLM-rec directions (RAG / GraphRAG, semantic IDs, LLM user towers, LLM-generated knowledge, LLM judges, multi-interest, PEFT) survive strict evaluation on news?
- **RQ4** At MIND-large scale: linear fusion or GBDT — and how trustworthy are validation estimates?
- **RQ5** Where is the remaining gap to fine-tuned PLMs, and does partial fine-tuning close it? (v17, pending)

Contributions

- A controlled harness: chronological split, test touched once, paired 95% CIs per impression, cold / warm slices, measured seed noise, pre-registered predictions, 73 automated tests.
- ≈ 16 technique families (≈ 150 variants) on MIND-small against retrained baselines; the V10 ↔ paper gap traced to pre-trained word vectors.
- Data findings: cold dominance, a warm validation day, the role of text fields, popularity as streaming feedback.
- Model findings: late interaction ≥ learned readers on small data; encoder size is not the bottleneck; GBDT saturates; user-fold CV is optimistic.
- A MIND-large pipeline on 1 × T4 / 32 GB with true peak-RAM accounting; honest positioning against published results.

3. LLM-RecSys 2024–2026: Seven Directions We Examined

Direction	Representative top-tier work	Tested here	Outcome on MIND
1. LLM as content encoder	LLM2Vec (COLM'24); Qwen3-Embedding (2025); LLMEmb (AAAI'25); LLM2Rec (KDD'25); LEARN (AAAI'25); PAD (SIGIR'25)	encoder sweep, concat encoder, learned readers (H8, H10)	aggregation & fields > encoder size
2. LLM-generated knowledge	KAR (RecSys'24); ONCE (WSDM'24); EmbSum (RecSys'24); IRLLRec (SIGIR'25); PNR-LLM (WWW'25 Comp.)	category / entity text (F); KAR zero-shot (H6); history-as-text (H1)	+0.0182 (Qwen3); KAR mixed
3. User modeling & multi-interest	MINER (ACL-F'22); CAUM (SIGIR'22); Iana et al. (SIGIR'23); attention masks for LLM users (2026)	LoRA user tower, 3 masks (H5); k-means & K-interest readers (H11)	not reproducible / negative
4. Generative rec & semantic IDs	TIGER (NeurIPS'23); LETTER (CIKM'24); HSTU (ICML'24); ActionPiece (ICML'25); RPG (KDD'25); OneRec (2025); GRID (CIKM'25)	RQ-KMeans 3×256 + SID-GPT slate ranker (H7)	≈ random (0.2898 vs 0.2897)
5. Reasoning	ReaRec (2025); OneRec-Think (ACL'26); SIDReasoner (KDD'26); FLR, VRec, list-wise alignment (SIGIR'26)	not tested (inference cost)	—
6. RAG / GraphRAG / agents	GraphRAG (2024); K-RagRec (ACL'25); LLM-agent survey (EMNLP-F'25)	representation RAG (H3); GraphRAG-lite + LLM summaries (H4)	≤ +0.004; sign flips val → test
7. LLM as ranker / judge	LLMRank (ECIR'24); LLM4Rerank (WWW'25)	Qwen3-1.7B Yes/No judge on top-10 (H9)	−0.0112 (n = 2,000)

** Venues verified by web search (Oct 2026); the complete reference list is in the report and on the References slide.*

3.1 Directions 1–2: LLM Encoders & LLM-Generated Knowledge

Direction 1 — LLMs as content encoders

- **LLM2Vec** (COLM'24): decoder-only LLM → encoder (bidirectional attention + contrastive training).
- **Qwen3-Embedding** (2025): instruction-aware, last-token pooling, 0.6B–8B.
- **LLMEmb** (AAAI'25), **LLM2Rec** (KDD'25): adapt the LLM with supervised / collaborative fine-tuning, then use it as an item-embedding model.
- **LEARN** (AAAI'25): frozen LLM item encoder + twin-tower bridge to the collaborative space (industrial).
- **PAD** (SIGIR'25): pre-train, align, disentangle text and collaborative views — largest gains on cold items.
- News: **Zhao et al.** (ECIR'25) — larger LMs are not reliably better on MIND.

Tested here: frozen BGE-small/base/large, Qwen3-Emb-0.6B and their concatenation, zero-shot and as reader input.

Direction 2 — LLM-generated knowledge

- **KAR** (RecSys'24): LLM reasoning + factual knowledge → hybrid-expert adaptor; deployed on Huawei news & music.
- **ONCE** (WSDM'24): open-source LLM as encoder + closed-source LLM for prompt augmentation (up to +19.32% rel.).
- **EmbSum** (RecSys'24): LLM-supervised user-interest summaries → poly-embeddings.
- **Yada & Yamana** (2024): GPT-4 category descriptions on MIND (up to +5.8% AUC).
- **PNR-LLM** (WWW'25 Comp.): hierarchical prompting to enrich news; **IRLLRec** (SIGIR'25): align LLM intents with interaction intents.

Tested here: category / entity names in the embedded text (F); KAR-style item expansion and reader profiles with Qwen3-1.7B (H6).

3.2 Directions 3–5: User Modeling, Generative Semantic IDs, Reasoning

Direction 3 — user modeling & multi-interest

- Multi-interest networks: Li et al. (CIKM'19), ComiRec (KDD'20); news: **MINER** (ACL-F'22), **CAUM** (SIGIR'22).
- **Iana et al.** (SIGIR'23): late fusion (dot products with clicked news) often beats complex user encoders.
- **Yuan et al.** (2026): attention masks for decoder-only user embeddings (causal / hybrid / bidirectional).

Tested here: LSE late interaction; k-means and K-interest readers; Qwen3 LoRA user tower with 3 masks.

Direction 5 — reasoning (2025–2026)

- **ReaRec** (2025) latent multi-step reasoning; **OneRec-Think** (ACL'26) interleaved text reasoning, deployed.
- **SIDReasoner** (KDD'26); SIGIR'26: **FLR**, **VRec** (reason–verify–recommend), Bayesian list-wise alignment.

Not tested here: per-impression inference cost (conditional future work).

Direction 4 — generative recommendation & semantic IDs

- **TIGER** (NeurIPS'23): RQ-VAE semantic IDs + autoregressive retrieval; **LETTER** (CIKM'24): learnable tokenizer.
- **HSTU** (ICML'24): generative recommenders with compute scaling laws.
- 2025: **ActionPiece** (ICML'25) context-aware tokenization; **RPG** (KDD'25) long semantic IDs in parallel; **OneRec** (Kuaishou) one generative model for retrieve + rank; **GRID** (CIKM'25) open framework.
- 2026: SID-level evaluation can overstate item-level accuracy through code collisions (Zhang et al.).

Tested here: RQ-KMeans 3 × 256 codes + a 4-layer SID-GPT used as a slate ranker (H7).

3.3 Directions 6–7, the News-PLM Line & Evaluation Rigor

Direction 6 — RAG, GraphRAG, agents

- **GraphRAG** (2024): community summaries over a knowledge graph; **K-RagRec** (ACL'25): KG retrieval for LLM recommenders.
- LLM-agent recommenders survey (EMNLP-F'25): recommender-, interaction- and simulation-oriented agents.

Tested here: entity / kNN RAG on article vectors; Louvain communities + Qwen3-1.7B summaries.

Direction 7 — LLM as ranker / judge

- **LLMRank** (ECIR'24): zero-shot ranking with order and popularity biases; **LLM4Rerank** (WWW'25).

Tested here: Qwen3-1.7B Yes/No log-odds judge re-ranking the top-10.

News recommendation with PLMs

- **PLM-NR** (SIGIR'21): PLM news encoder, fine-tune the last 2 layers; **UNBERT** (IJCAI'21); **GLORY** (RecSys'23).
- Training cost is the bottleneck: **SpeedyFeed** (KDD'22, > 100× speed-up), **GRAM** (NAACL'22), **Tiny-NewsRec** (EMNLP'22).
- **EB-NeRD** (RecSys Challenge'24) ships article bodies (MIND does not).

Evaluation rigor

- Weak baselines inflate progress (**Dacrema et al.**, RecSys'19); a reproducibility study finds LLMRec behind established baselines (**Fioretti et al.**, RecSys'25).

→ Our protocol: retrained baselines, test touched once, paired CIs, measured seed noise.

4. Data: MIND-small vs MIND-large (measured from run logs)

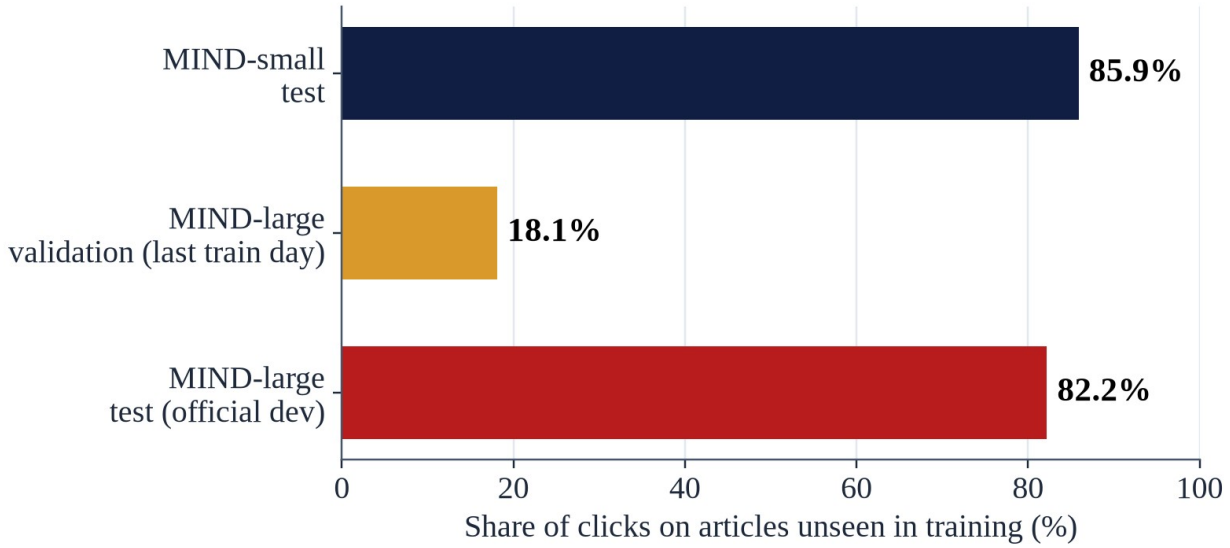
Statistic	MIND-small	MIND-large
Articles (train + dev merged)	65,238	104,151
Official train impressions	156,965	2,232,748
train_core / validation (last day)	126,695 / 30,270	1,801,231 / 431,517
Validation actually used	30,270	60k (v14); 150k (v15+)
Test impressions (official dev)	73,152	376,471
Training samples (click + 4 negatives)	—	2.71 M (0.6 M v14; 1.2 M v15+)
Categories / subcategories	18 / 270	—
Title / abstract length (words)	10.7 / 34.9	—
Articles with entity names	87.5% (2.4 / article)	—
Test readers' history length	mean 32.3; 19.6% > 50	—
Test clicks on cold articles	**85.9%**	**82.2% (val. 18.1%)**
Article body text	not provided	not provided

Reading the table

- Both test sets are overwhelmingly cold; the MIND-large validation day is not.
- Text is short and bodiless → field and aggregation choices matter.
- Most readers have < 50 clicks → longer histories barely help (+0.001).
- MIND-large forced a lean loader, subsampled validation / training and a RAM-aware pipeline (Section 4.3).

** Counts printed by the data audit and loaders of runs v13 (small) and v14–v16b (large).*

4.1 Finding 1: Test Clicks Are Cold, the Validation Day Is Warm



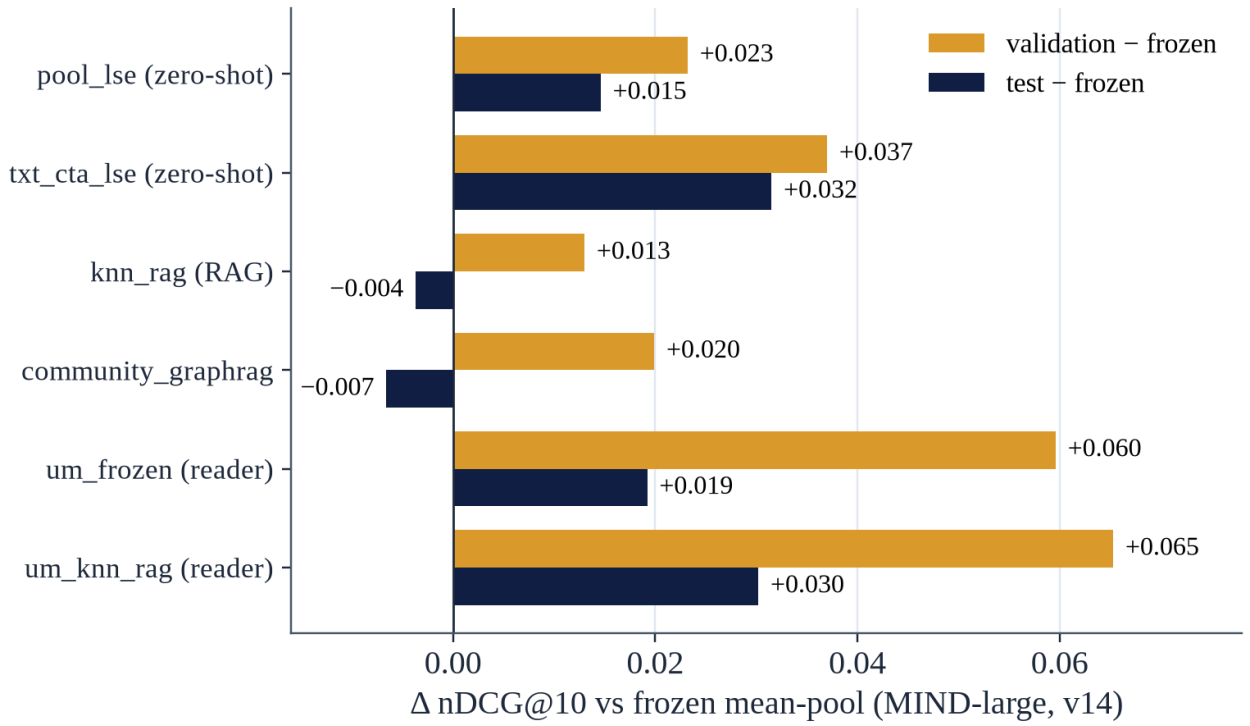
What we measured

- Cold = the clicked article appears in no training row (history or impression).
- Test: 85.9% (small) and 82.2% (large) of clicks are cold; MIND-large validation: 18.1%.

Why it matters

- On warm articles click-trained models win (warm AUC 0.56–0.64 vs 0.44–0.54 for zero-shot embeddings).
- Frozen title + abstract embeddings have warm AUC < 0.5 (BGE-small 0.4416): similarity to the history anti-correlates with clicks on seen articles.
- Fix (v16): select fusion weights on a cold-weighted validation nDCG@10 (80% cold).

4.2 Finding 2: A Warm Validation Day Misleads Model Selection (v14)



Gain over frozen: validation vs test

- Zero-shot scorers keep most of their gain (test − validation: −0.005 … −0.009).
- Learned readers lose 0.035–0.040 (um_knn_rag +0.065 → +0.030).
- **kNN-RAG (+0.013 → −0.004) and GraphRAG (+0.020 → −0.007) flip sign.**

Consequence

- Adding 3 readers + a 100-click history to the fusion raised validation (0.4152 → 0.4177) but lowered test: −0.0014 [−0.0017, −0.0011].
- Every row is "easier" on test (even random vectors +0.020): compare deltas against a control, never absolute levels across splits.

4.3 Findings 3–5: Text Fields, Popularity and Memory

Text fields matter (v13, MIND-small)

- Dropping the abstract: −0.016 to −0.032 nDCG@10.
- Category / subcategory in the text: **Qwen3 +0.0182** [+0.0169, +0.0196], validation agrees; BGE-small +0.0013 (validation disagrees). Same on MIND-large: Qwen3 +0.0186, BGE-large −0.0030.
- The gain sits in the warm slice (warm AUC 0.4939 → 0.6180; cold +0.0005).
- Entity names +0.0033; 100 / 200-click histories ≈ +0.001.

Popularity: strong but streaming

- Online popularity alone: 0.4026 (small) / 0.4040 (large) nDCG@10 — above every V10 neural model; it reads earlier clicks of the test day → kept in a separate "+pop" frame.

Scale & memory (MIND-large)

- True peak RAM 21.2 GB (stacker stage; 32 GB container; GPU 5.0 GB), measured from v16b on.
- v15's "16.2 GB" was an end-of-stage snapshot, not a peak.
- v16 died after 1.6 h: the kNN-view gather allocated 2 × [104,151 × 20 × 2,048] float32 = 34.1 GB.
- Chunked computation (≈ 600 MB blocks): identical results (< 1e-6), +0.46 GB peak.
- Free Colab (~12.7 GB) is not enough at this scale; ≥ 24 GB is needed.

5. Evaluation Protocol & Leakage Controls

Chronological split, test touched once

- Validation = last day of MIND_train; train_core = earlier days; MIND_dev = test. Every choice (τ, epochs, fusion weights, stacker configurations) uses validation only.

Statistics

- Paired 95% CI per impression: $\bar{\Delta} \pm 1.96 \cdot SE$ with impression-clustered SE (analytic, not bootstrap); subset rows are compared on the same impressions; Bonferroni across technique families.
- Seed noise measured: 0.007–0.014 nDCG@10 between two seeds of one recipe — far wider than the CI half-width (± 0.002).

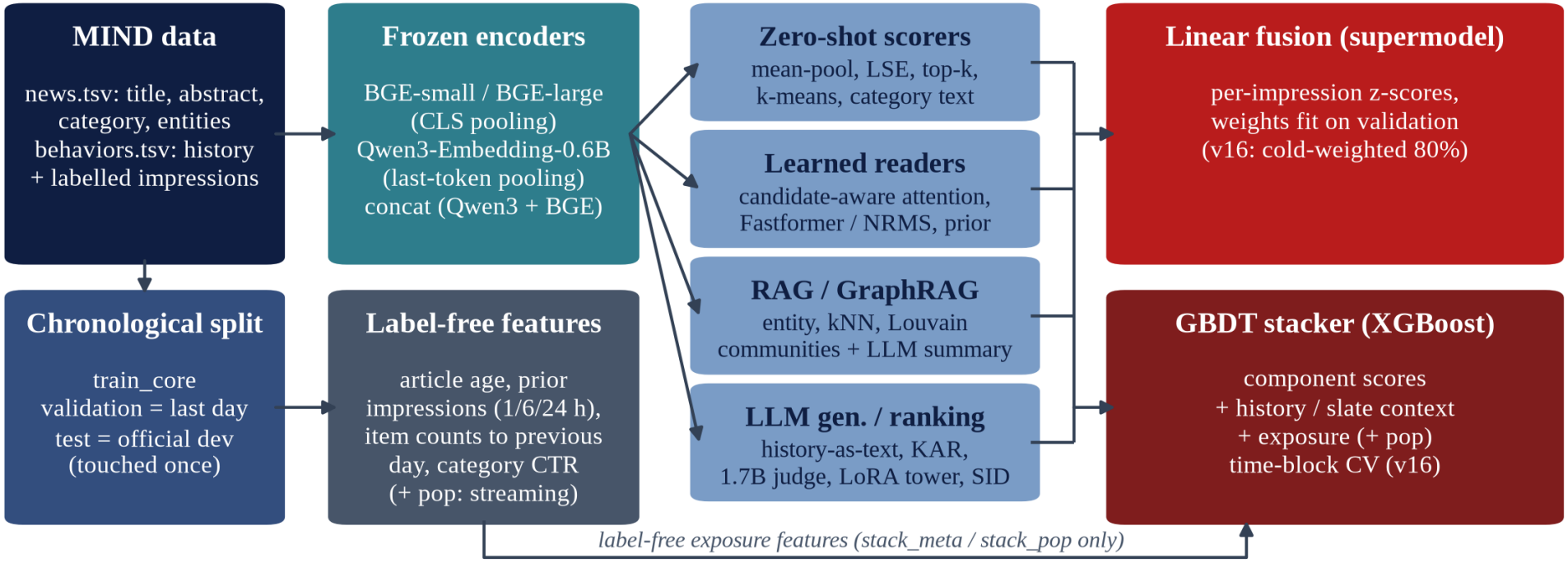
Reproduction checks against the original V10 benchmark

- BGE-small mean-pool = V10 (AUC 0.6241 / nDCG@10 0.3909); popularity and frozen+pop match; NRMS rung 0 reproduces V10 (copied weights: $|\Delta\text{score}| \leq 3 \cdot 10^{-8}$); random vectors give AUC 0.50.

Leakage tests (73 automated tests on CPU)

- Shuffling test labels leaves stack_llm / stack_meta predictions bit-identical; flipping one fold's labels leaves that fold's out-of-fold scores unchanged; two deliberate leaks are caught. Predictions are written down before every run (Section 9.1).

5.1 System Architecture



Evaluation: AUC · MRR · nDCG@5/10 | paired 95% CI per impression | cold / warm slices | measured seed noise | V10 reproduction checks

** Encoders are frozen in v7–v16b; only heads, readers and the stacker are trained. v17 unfreezes the last 2 layers of BGE-large.*

5.2 Linear Fusion and the GBDT Stacker

Linear fusion (supermodel)

- Each component exports raw scores; z-scored inside each impression.
- s_super = z_base + Σk wk·zk; base = best single component on validation (weight 1); wk ∈ {0, …, 4} by coordinate ascent; test touched once per rung.
- Ladder: super_base → super_zs (+ zero-shot) → super_llm (+ readers) → super_llm+pop.
- v16: base and weights chosen on cold-weighted validation nDCG@10 (80% cold).

Stacker variants

- stack_llm = scores + ctx; stack_meta = + meta (label-free, reads impression-log exposure); stack_pop = + pop (streaming).

GBDT stacker (XGBoost, GPU)

- Trained on validation impressions only; test score = mean z-scored prediction of the fold models.
- **scores**: z-scores of every exported component (Qwen3, BGE-large, concat) + a few raw values.
- **ctx**: 11 history / slate features from frozen vectors (+ 4 IDF entity-overlap features in v16).
- **meta** (16): item counts up to the previous day, smoothed CTR, category CTR, article age, prior impressions, 1/6/24-h exposure — no labels.
- **v15**: 4 user folds; objective chosen on fold 0 (binary:logistic); eta 0.06, depth 7, ≤ 800 trees.
- **v16**: time-block folds, selection on new-article impressions, 5 configurations on a forward holdout, blend of the best 2; the v15 recipe kept as a paired control.

6. Techniques Tested: Catalogue and Verdicts

Code	Direction	What we implemented	Verdict (Δ nDCG@10 vs own control)
H1	History as text	10 latest titles → one instructed query, same encoder	−0.0211 … −0.0014
H2	Co-click adaptation	residual head, LoRA / DoRA, Matryoshka, impression hard negatives	+0.0016 … +0.0029; cold AUC ↓
H3	Representation RAG	article vector + Wikidata entities or k nearest past articles	−0.0080 … −0.0003
H4	GraphRAG-lite	Louvain communities; community summaries by Qwen3-1.7B	+0.0043 (cold AUC −0.0031)
H5	LLM user tower	Qwen3 + LoRA, InfoNCE; causal / bidirectional / soft masks	mask effects not reproducible
H6	KAR-style knowledge	Qwen3-1.7B item expansion / reader profiles (n = 2,500)	+0.0079 / −0.0308
H7	Semantic IDs	RQ-KMeans 3 × 256; prefix profile; SID-GPT slate ranker	+0.0035; SID-GPT ≈ random
H8	**Late interaction**	**log-sum-exp, top-k, MaxSim, recency pooling**	**+0.0128 (LSE)**
H9	LLM judge	Qwen3-1.7B Yes/No log-odds on the top-10 (n = 2,000)	−0.0112 [−0.0216, −0.0008]
H10	Learned readers	candidate-aware attention on frozen vectors; Fastformer / NRMS readers	< LSE (small data); > LSE (large)
H11	Multi-interest	k-means interests (K = 2, 3, 5); K-query readers ± diversity	−0.0030 vs LSE; no gain
F	**More data fields**	**category, entity names, separate views, 100 / 200 clicks**	**+0.0182 (Qwen3 category)**
R	Retrained baselines	NRMS / NAML / Fastformer, GloVe-300, published recipes, refit	NRMS refit 0.4078
S	Fusion & stacking	z-score linear fusion; XGBoost stacker (llm / meta / pop)	Section 8
FT	Partial fine-tuning	last 2 layers of BGE-large under the reader + frozen twin (v17)	pending

7. MIND-small: Retrained Baselines vs Zero-shot LLM Embeddings

Model (content only)	AUC	MRR	nDCG@5	nDCG@10	cold AUC
NRMS — V10 recipe (old)	0.6181	0.2809	0.3020	0.3685	—
NRMS — published recipe, GloVe	0.6392	0.3037	0.3293	0.3937	0.6471
NAML — published recipe	0.6471	0.3079	0.3364	0.4006	0.6698
Fastformer — mean of 2 seeds	0.6426	0.3000	0.3289	0.3927	0.6623
NRMS refit (+ validation days)	**0.6555**	**0.3140**	**0.3443**	**0.4078**	**0.6700**
BGE-small mean-pool (zero-shot)	0.6242	0.3002	0.3299	0.3909	0.6617
BGE-small + LSE	0.6386	0.3118	0.3427	0.4037	0.6675
Qwen3-Emb-0.6B + LSE + category	**0.6556**	**0.3201**	**0.3509**	**0.4094**	**0.6637**
Published NRMS (Iana et al., SIGIR'23)	0.6658	0.3144	0.3499	0.4121	—

RQ1 — answer

- Retrained baselines beat the old V10 numbers by up to +0.039 nDCG@10; NRMS refit sits 0.004 below the published value.
- Zero-shot BGE-small ≈ a one-seed baseline; late interaction lifts it to 0.4037.
- **Best zero-shot 0.4094 vs NRMS refit 0.4078: +0.0016, bounding CI [−0.0023, +0.0055] includes 0;** same AUC, +0.006 MRR / nDCG@5, lower cold AUC.

Safe claim: frozen LLM embeddings match trained baselines without training or IDs — they do not beat them.

** Seed noise of a single trained model: 0.007–0.014 nDCG@10 (Section 7.2). Published row: mean of 5 runs.*

7.1 MIND-small: What Actually Helps (and What Does Not)

Change (vs its own control)	Δ nDCG@10 [95% CI]	Val. agrees?
Late interaction LSE (BGE, vs mean-pool)	**+0.0128**	**yes**
Late interaction top-k (BGE)	+0.0107	yes
Category in text — Qwen3 (vs LSE)	**+0.0182 [+0.0169, +0.0196]**	**yes**
Category in text — BGE-small (vs LSE)	+0.0013 [+0.0000, +0.0026]	no
+ entity names — BGE (ctae vs cta)	+0.0033 [+0.0025, +0.0041]	yes
200-click history — BGE (vs 50)	+0.0012 [+0.0008, +0.0016]	tiny
Title only — Qwen3 (drop abstract)	−0.0321 [−0.0338, −0.0304]	yes
Encoder: BGE-large vs BGE-small	+0.0015 (n.s.)	—
Encoder: Qwen3-0.6B vs BGE-small	−0.0125	—
Encoder: TF-IDF + LSA vs BGE-small	−0.0443	—

RQ2 — answer

- The bottleneck of frozen-embedding recommenders is **history aggregation** and **which fields reach the encoder**, not encoder size.

Late interaction (no parameters)

$s_{LSE}(u, c) = \tau \cdot [\log \Sigma_h \exp(\langle v_h, v_c \rangle / \tau) - \log L]$

- $\tau \to 0$: MaxSim; $\tau \to \infty$: mean-pool; τ chosen on validation.
- Replicated on Qwen3: +0.0127.

7.2 Why the Old Baselines Were Weak: Ablation Ladder & Seed Noise

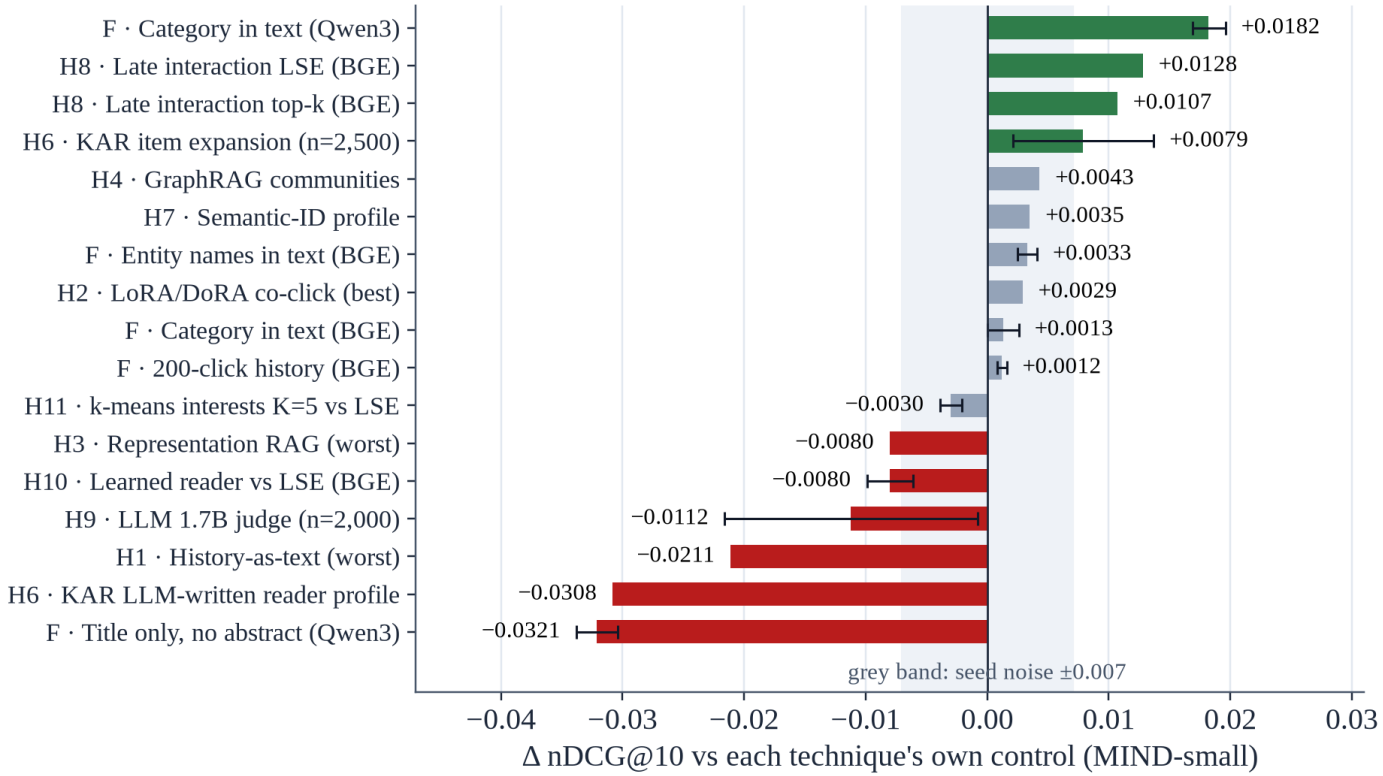
NRMS ablation rung (cumulative)	AUC	nDCG@10	Δ nDCG@10 vs previous [95% CI]
Rung 0 — V10 recipe in the new harness	0.6156	0.3694	—
Rung 1 — + title-only, UNK, padding mask, negatives re-drawn every epoch	0.6102	0.3661	−0.0033 [−0.0051, −0.0015]
Rung 2 — + standard size / lr, random vectors (not converged)	0.5818	0.3447	−0.0213 [−0.0229, −0.0197]
NRMS, published recipe — + GloVe-300	**0.6392**	**0.3937**	**+0.0490 [+0.0471, +0.0509]**
NRMS refit — + training on the validation days	0.6555	0.4078	+0.0141 [+0.0126, +0.0156]

Same recipe, two seeds	seed A	seed B	gap
NRMS (published recipe)	0.3937	0.4005	0.0068
Qwen3 LoRA user tower (causal)	0.3944	0.3820	0.0124
Qwen3 reader um_frozen	0.3929	0.4063	0.0134
Qwen3 reader um_text	0.4173	0.4036	0.0137

Take-aways

- Pre-trained word vectors (GloVe) explain most of the V10 ↔ paper gap; tokenizer / mask / negatives do not.
- A 'weak' baseline creates a 0.04 gap — larger than any LLM effect we measured.
- Seed noise ≫ paired CI: single-seed gaps < 0.007 are not findings.

7.3 Null and Negative Results Across LLM Directions (MIND-small)



Negative or ≈ 0

- Representation RAG −0.0080 … −0.0003; GraphRAG +0.0043 (cold AUC −0.0031).
- SID-GPT slate ranker 0.2898 ≈ random vectors 0.2897.
- LLM-written reader profiles −0.0308; history-as-text −0.0211 … −0.0014.
- Qwen3-1.7B judge −0.0112 [−0.0216, −0.0008]; fused with the base score −0.0051 (n.s.).
- LoRA / DoRA +0.0016 … +0.0029 with lower cold AUC.

RQ3 — answer

- Only the cheap form of LLM knowledge (category text) and late interaction give robust gains on news.

7.4 Multi-interest and the LLM User Tower: No Robust Gain

Enc.	Reader (test nDCG@10, mean of seeds)	nDCG@10	std	interest cos
BGE	um_frozen (1 vector, 3 seeds)	0.3968	0.0016	—
BGE	mi1 (K = 1 control)	0.3906	0.0035	—
BGE	mi4 (K = 4)	0.3944	0.0011	0.85–0.87
BGE	mi4d (K = 4 + diversity penalty)	0.3972	0.0055	0.34–0.39
Qwen3	um_frozen (2 seeds)	0.3996	0.0067	—
Qwen3	mi1 (K = 1 control)	0.4008	0.0018	—
Qwen3	mi4 (K = 4)	0.4029	0.0019	0.91
Qwen3	mi4d (K = 4 + diversity penalty)	0.3965	0.0032	0.38–0.52

- k-means interests sit between mean-pool and LSE, monotone in K (−0.0079 / −0.0048 / −0.0030 vs LSE).
- Without a penalty the K interests collapse (cos 0.85–0.91); with it they separate but ranking does not improve.

Qwen3 + LoRA tower mask	nDCG@10	+ meanpool
causal (seed 0)	0.3944	0.4026
bidirectional	0.3854	0.3965
soft (v7)	0.3964	0.4043
soft (v13 re-run)	—	0.3956
causal (seed 1)	0.3820	0.3923

- soft − causal: +0.0020 (v7) → −0.0063 (v13), same code and seed.
- Seed gap 0.0124 ≥ any mask effect → no evidence that masks matter here (n = 30,000).

8. MIND-large (v15): Components, Linear Fusion and GBDT Stacker

Row	AUC	MRR	nDCG@10	cold AUC	warm AUC
Qwen3 frozen (mean-pool)	0.6215	0.2922	0.3787	0.6583	0.4784
BGE-large frozen (mean-pool)	0.6297	0.3042	0.3933	0.6632	0.4985
super_llm (linear fusion)	0.6774	0.3286	0.4268	0.7004	0.6207
stack_llm (GBDT: scores + context)	**0.6843**	**0.3355**	**0.4332**	**0.7003**	**0.6534**
stack_meta (+ label-free exposure)	**0.6998**	**0.3500**	**0.4508**	**0.7280**	**0.6187**
frozen + pop (streaming)	0.6757	0.3324	0.4312	0.7259	0.5098
super_llm + pop (streaming)	0.7114	0.3574	0.4599	0.7446	0.6192
stack_pop (streaming)	0.7206	0.3680	0.4730	0.7524	0.6313

Paired comparisons (nDCG@10)

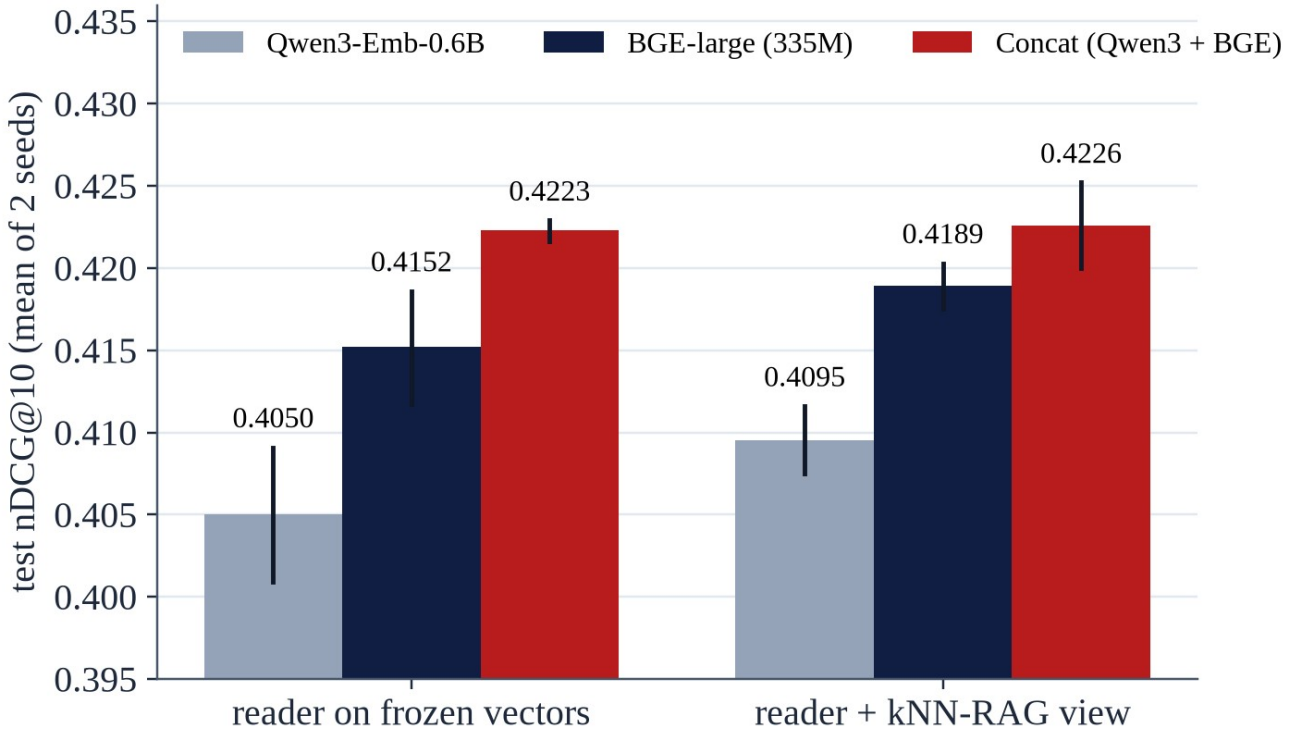
- stack_llm − super_llm: +0.0064 [+0.0059, +0.0069], but cold AUC −0.0001: the gain is on seen articles.
- stack_meta − stack_llm: +0.0176; cold AUC +0.0277, warm AUC −0.035.
- stack_pop − super_llm+pop: +0.0131.

Caveat

- Stacker out-of-fold validation sat 0.03–0.04 above test (0.4909 → 0.4508) — tested in v16b (Section 8.3).

** MINDlarge_dev, 376,471 impressions; stack_meta reads exposure in the impression log (no labels); the last three rows are streaming.*

8.1 Readers by Encoder (v16b): A Bigger LLM Is Not Better

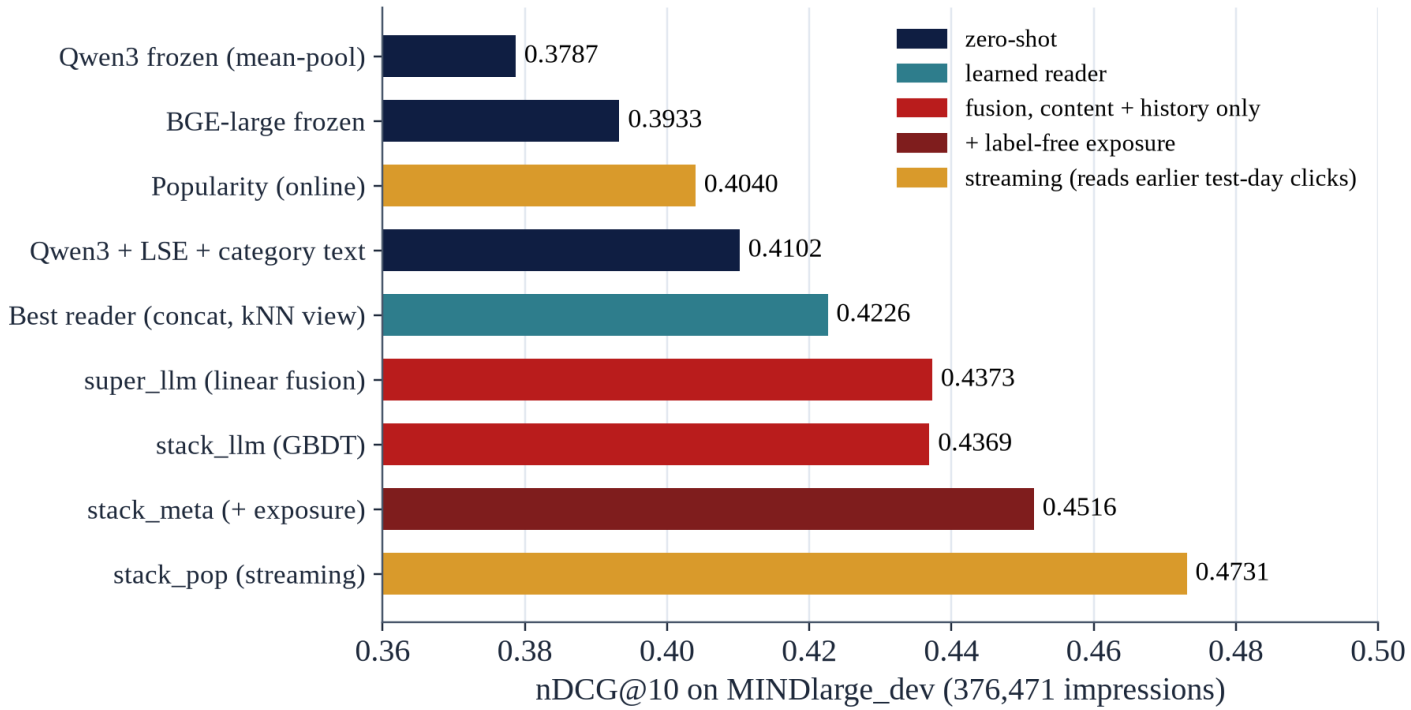


Same 128-d head, 2 seeds each

- BGE-large (335M) − Qwen3-Emb (0.6B): **+0.0102** [+0.0095, +0.0108].
- Concat − BGE-large: +0.0071 [+0.0066, +0.0076]; ordering holds for both seeds.
- kNN-RAG view: +0.0037 on BGE-large; +0.0003 on concat (n.s., cold AUC +0.0088).
- With 1.2 M training samples a reader beats LSE on the same encoder (BGE-large +0.0105) — the reverse of MIND-small (−0.0080).

Diversity of representation spaces helps more than size (consistent with Zhao et al., ECIR'25).

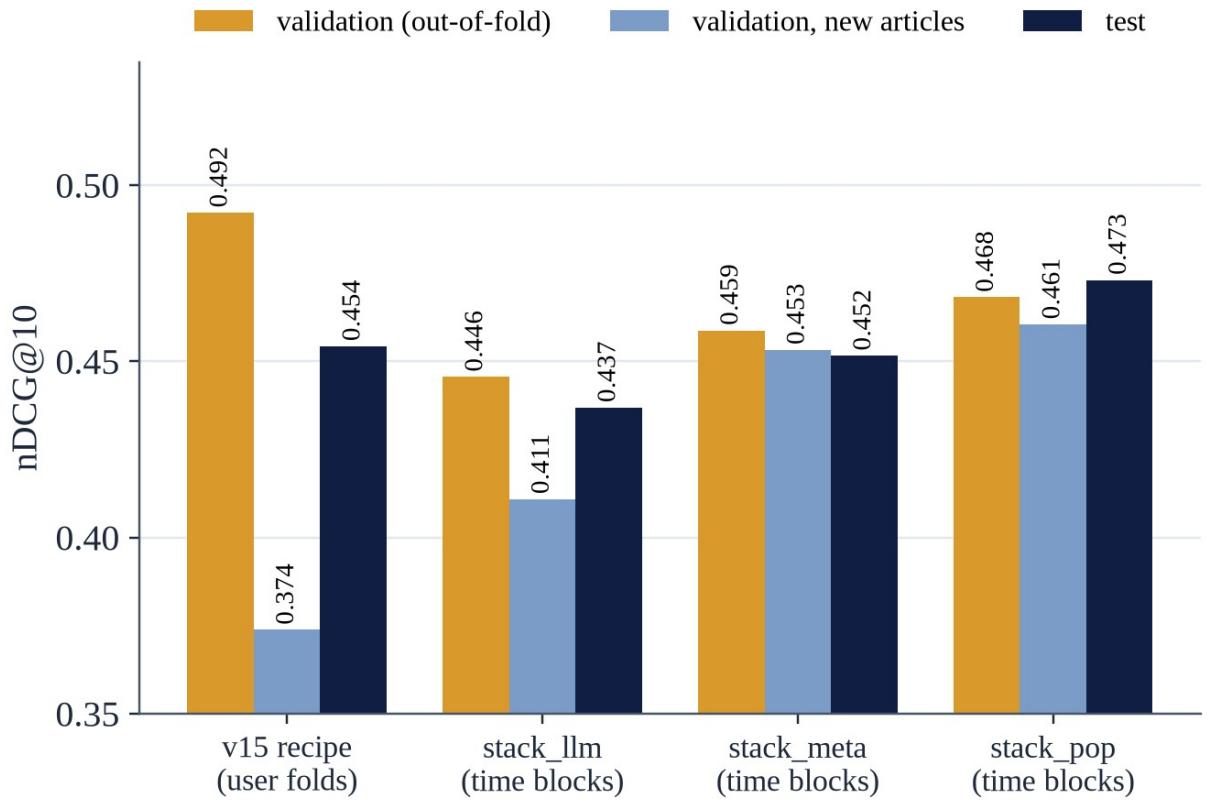
8.2 MIND-large Results Ladder (nDCG@10)



Reading the ladder

- Content + history ceiling ≈ 0.437: linear fusion and GBDT tie.
- Label-free exposure adds +0.0143 (stack_meta).
- Streaming popularity adds +0.0215 more (stack_pop) — not comparable with offline papers.
- Colours mark the information each row uses, not the model family.

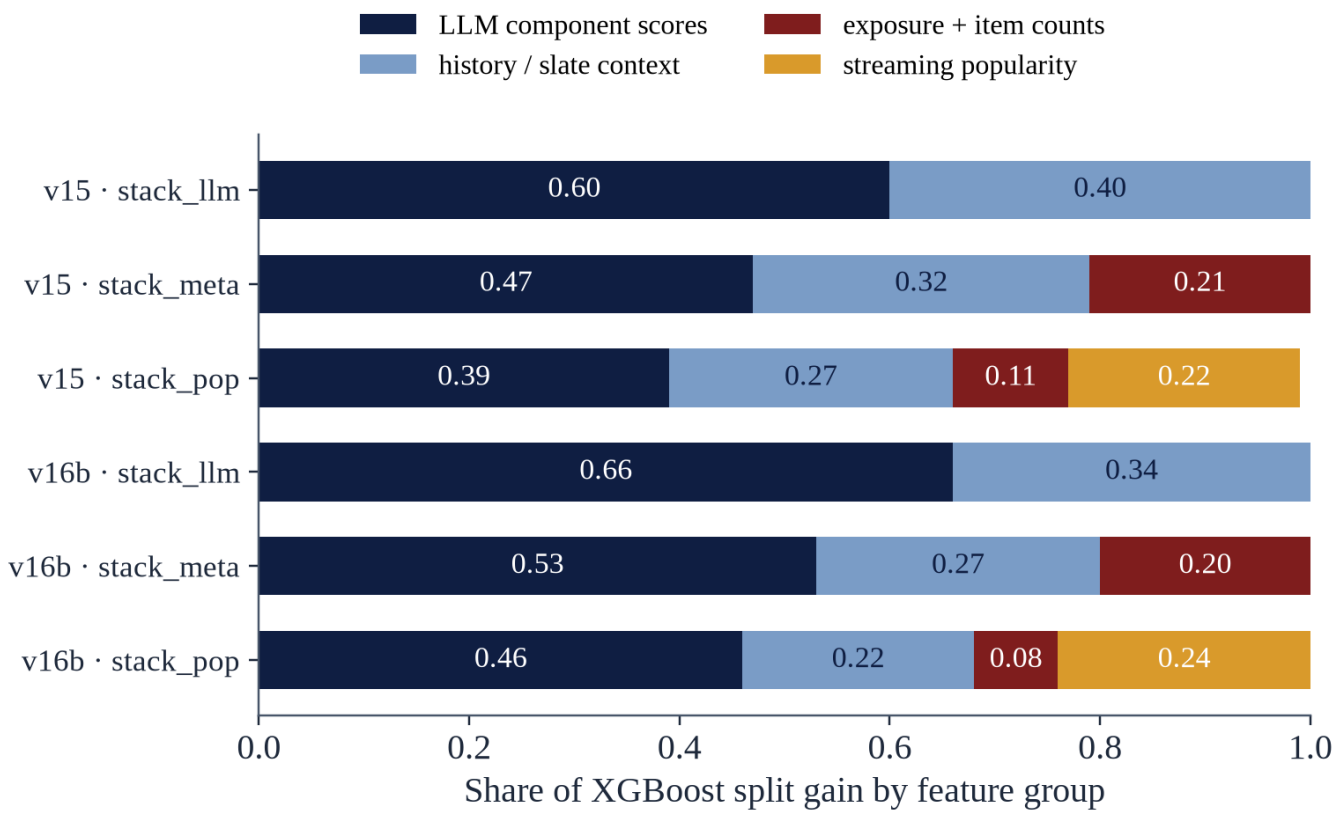
8.3 Stacker Saturation and Honest Cross-Validation (v16b)



v16b row	AUC	nDCG@10	v15
super_llm	0.6863	0.4373	0.4268
stack_llm	0.6853	0.4369	0.4332
stack_meta	0.7013	0.4516	0.4508
stack_meta_v15cfg (control)	0.7029	0.4543	—
stack_pop (streaming)	0.7196	0.4731	0.4730

- **Saturation:** content improved +0.0105, stack_meta only +0.0008; stack_llm − super_llm = −0.0004 [−0.0010, +0.0002].
- **Honest CV:** user folds overstate test by 0.038; time-block folds + new-article slice by 0.007.
- Yet the honest recipe is not better on test (−0.0027 [−0.0031, −0.0023] vs the v15 control): the *estimate* was wrong, not the model.

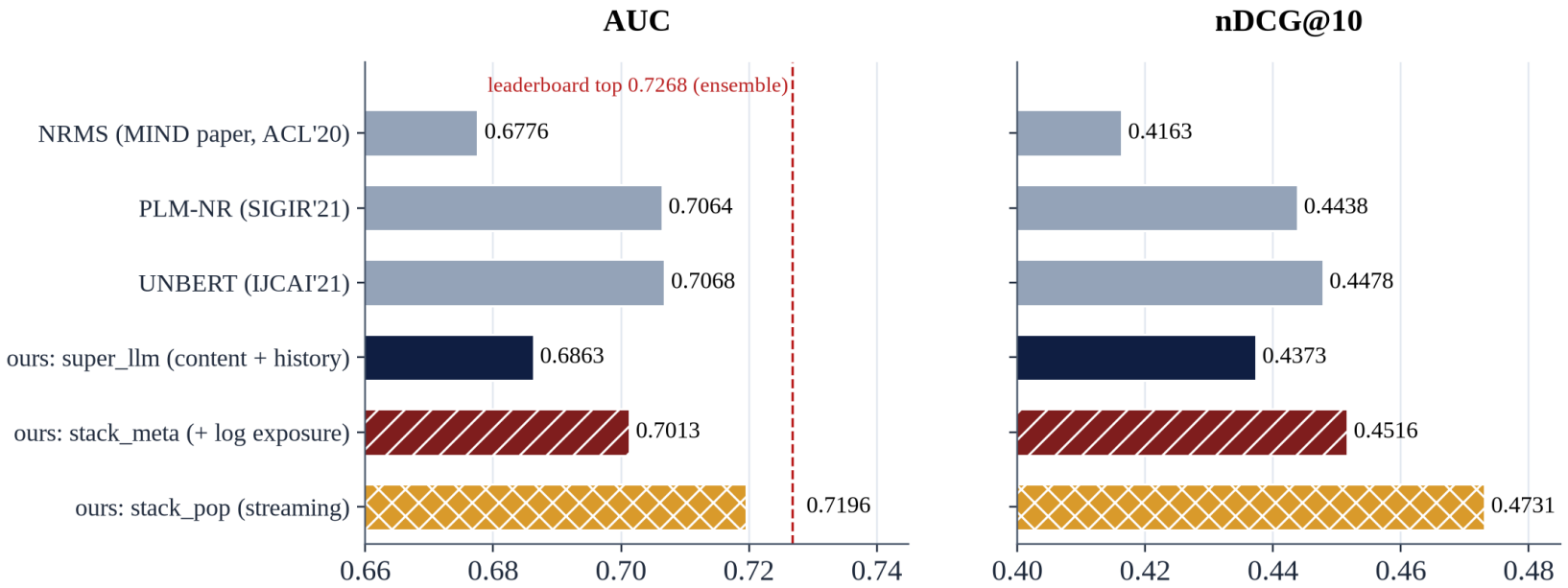
8.4 What the Trees Use: Split-Gain Shares



Interpretation

- LLM component scores: 0.47–0.66 of total split gain in stack_llm / stack_meta (0.39–0.46 in stack_pop).
- The context group (0.22–0.40) mixes LLM-derived similarities with non-LLM features — slate size (ctx_n_cand) alone takes 0.10–0.25.
- So the LLM share is not 'scores + context'; an '80% LLM' claim would overstate it.
- Top v16b features: cat:um_knn_rag (0.09–0.13), ctx_n_cand, bge:um_frozen, ctx_slate_max, m_ctr; streaming: pop, popw_ctr_1h.

9. Positioning Against Published MIND-large Results



- Content + history rows (super_llm, stack_llm: AUC 0.684–0.686) lie inside the NRMS range reported in the literature (AUC 0.670–0.691) and ≈ 0.02 AUC below PLM-NR (0.7064) / UNBERT (0.7068).
- stack_meta is within 0.005 AUC of PLM-NR and above it on nDCG@10 (0.4516 vs 0.4438), but reads impression-log exposure that PLM-NR / UNBERT do not use → a different information set; stack_pop needs test-day labels and cannot be submitted.

** Published: leaderboard test set, secondary sources (PDFs not opened). Ours: MINDlarge_dev. ~0.01 AUC differences are not model claims.*

9.1 Pre-registered Predictions vs Outcomes

Prediction (written before the run)	Measured	Outcome
v13: category is the best new field, $\|\Delta\| \lesssim 0.01$	Qwen3 +0.0182; BGE ≈ 0	direction right, size wrong
v13: k-means interests between mean-pool and LSE, possibly ≥ LSE	between, always < LSE	half right
v15: stack_llm > super_llm, Δ in +0.003 … +0.02	+0.0064	right
v15: stacker cold AUC > supermodel cold AUC	−0.0001 (stack_llm)	wrong
v15: stack_meta > stack_llm (~60%)	+0.0176	right
v16b: stack_meta AUC 0.698–0.712, nDCG@10 0.449–0.464	0.7013; 0.4516	inside
v16b: stack_llm gains ≥ +0.01 over v15 (~15–20%)	+0.0037	wrong
v16b: test − validation gap of stack_meta < control (~65%)	−0.0072 vs −0.0379	right
v16b (implicit): stack_meta > stack_meta_v15cfg	−0.0027	wrong
v17: PASS ~35% / INCONCLUSIVE ~25% / FAIL ~40%	not run yet	—

** Predictions were written in the README before each run; several of them turned out wrong, which is reported as is.*

10. Compute, Memory and Reproducibility

Run	Data	What changed	Time / resources
V10	small	original benchmark: NRMS / NAML / Fastformer, BGE zero-shot, popularity	—
v7	small	≈ 16 families / ≈ 150 variants; retrained baselines; ablation ladder	7.52 h (1 × T4)
v13	small	text fields, 100 / 200 clicks, multi-interest, Fastformer × 2 + refit	8.22 h
v14	large	one Qwen3 model + all techniques; linear supermodel	2.90 h
v15	large	GBDT stacker (XGBoost GPU), exposure features, BGE-large	2.73 h
v16	large	time-block CV, BGE-large / concat readers	died at 1.6 h (out of RAM)
v16b	large	chunked kNN view, true peak RAM, 128-d heads for 3 encoders	4.49 h; peak 21.2 GB
v17	large	gated fine-tuning of BGE-large's last 2 layers	est. 6–6.5 h (pending)

Engineering notes

- Each run is one 'Run All' Kaggle notebook (Tesla T4 16 GB, 32 GB container).
- 73 automated CPU tests before each run (correctness, leakage, memory).
- Exported component scores + resume allow a stacker-only re-run without re-encoding the catalogue.
- Serving latency was **not** measured in this work.

11. Answers to RQ1–RQ5 and Limitations

Answers

- **RQ1** Zero-shot ≈ retrained baselines (0.4094 vs 0.4078, n.s.).
- **RQ2** History aggregation and input fields, not encoder size.
- **RQ3** Most directions ≤ 0.005 or negative; weak baselines and a warm validation day create false positives.
- **RQ4** Linear fusion ≈ GBDT on content; GBDT adds value only via label-free exposure and saturates ≈ 0.45; time-block CV is needed.
- **RQ5** The ≈ 0.02 AUC gap to PLM-NR points to end-to-end fine-tuning → v17 (pending).

Limitations / threats to validity

- One dataset family and one test day; dev instead of the leaderboard test; published numbers are secondary.
- Many rows single-seed; CIs exclude training noise (0.007–0.014).
- No tuning beyond the stated grids; GloVe-6B; Fastformer re-implemented.
- LLM generation / judging only at 1.7B; Qwen3-Embedding 4B / 8B not tested; subset runs (n = 2,000–30,000).
- The 80% cold target used knowledge of the test distribution (not labels).

12. Next Step: Gated Fine-tuning Experiment (v17)

Design (PLM-NR recipe + SpeedyFeed-style caching)

- Cache layer-22 hidden states of BGE-large for 104,151 articles ([104,151 × 96 × 1,024] fp16 ≈ 20 GB on local disk); check cos ≥ 0.99 against the full encoder.
- Train the last 2 layers + reader (fp16 autocast); candidate-side gradients; history table refreshed every 600 steps.
- Two matched arms from the same start: unfrozen (ft_reader) vs frozen twin (ft_base) — same steps, samples and order.
- Why the twin: in a toy run, continued reader training alone gave +0.0215 vs +0.0040 for unfreezing.

Decision rule (cold-weighted validation)

- **PASS**: unfrozen − frozen ≥ +0.006 and raw mean not down > 0.001 → scale up (more layers / steps, both sides).
- **INCONCLUSIVE**: ≥ +0.002 → add seeds first.
- **FAIL**: otherwise → keep frozen readers; write the thesis on v16b.
- If PASS (predicted): ft_reader 0.418–0.428 test nDCG@10 vs 0.4152 for the frozen BGE reader.

PASS · ~35%
≥ +0.006 → full fine-tuning run

INCONCLUSIVE · ~25%
≥ +0.002 → more seeds before deciding

FAIL · ~40%
keep frozen readers; thesis on v16b

** Gate and probabilities were written in the README before the run; only validation decides.*

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** The complete, verified list (62 entries) is in the accompanying report.*

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XIN TRÂN TRỌNG CẢM ƠN!

Kính mong nhận được ý kiến trao đổi học thuật của Quý Thầy Cô và các bạn!

TÓM LƯỢC BA LUẬN ĐIỂM KHOA HỌC CỐT LÕI:

- Data decide the evaluation: 82–86% of test clicks are cold and the validation day is warm — ignoring this selects the wrong models.
- Frozen LLM embeddings match, not beat, properly trained baselines; robust gains come from history aggregation and input fields, not encoder size.
- GBDT stacking adds value only through label-free exposure features and saturates (≈ 0.45 nDCG@10); the remaining gap is end-to-end fine-tuning (v17).

Người thực hiện: NGUYỄN NAM HÁN · MSHV: 20252095M
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